# IMT Doctrine v0.2.0-demo-1-g11bc310

***Release v0.2.0-demo-1-g11bc310***

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**Doctrine working group**

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# DOCTRINE


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**Warning**

**Demonstration repository.** This site shows how the LandSAR IMT doctrine could be written, reviewed and published from Git. Structure follows the draft; text is short illustrative placeholder content, not agreed doctrine.

**Cite as:** IMT Doctrine, version v0.2.0-demo-1-g11bc310.

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CHAPTER

ONE

# PART 1 FOUNDATIONAL OVERVIEW

## 1.1 1.1 Introduction

LandSAR volunteers provide an independent, impartial and professional response in support of New Zealand's two Coordinating Authorities: New Zealand Police and the Rescue Coordination Centre New Zealand (*RCCNZ*). When a Coordinating Authority engages LandSAR, specialist Incident Management Teams (*IMT*s) manage the operation. Volunteers also provide search and rescue capability during civil defence emergencies.

Land search management has developed unevenly, drawing on many government and non-government sources, so practice varies across the country. This doctrine sets out the principles, core concepts and proven methods that govern how IMTs organise, train and operate. It is the single source of truth for "how we do things".

The doctrine is:

- **Integrated**: it fits within the legislative and operational requirements of the Coordinating Authorities.
- **Authoritative, not prescriptive**: it gives a decision-making framework while preserving the flexibility IMTs need.
- **Based on proven practice**: it captures lessons from past operations and reviews.
- **Grounded in search theory**: local, national and international.
- **The foundation for consistency**: shared terms, processes and expectations so groups, Police and partners work together seamlessly.
- **Built for continuous learning**: one place to add, amend and improve.

## 1.2 1.2 Purpose

To set a unified, nationally consistent approach to IMT response processes and practices. The doctrine:

- draws on input from IMTs nationwide
- brings together national and regional coordinators, tutors, group IMT leaders and Police SAR coordinators
- pairs a single source of truth with a feedback loop, so operational learnings update the document
- embeds its principles in IMT training and operational evaluation
- makes it simple to benchmark against international practice.

## 1.3 1.3 Scope

The doctrine covers the delivery of effective land search and rescue within an operational management context. It cascades from principles to fundamental concepts, then to operational application, guidelines, and the templates, checklists and forms that follow from them. Training material is built on the same foundation.

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It is not:

- a guide to every SAR sector function or agency activity
- a prescriptive, step-by-step operations manual
- a set of local standard operating procedures.

**Note**

Scope is still to be settled against the doctrine brief, for example whether notification and Coordinating Authority processes are in or out.

## 1.4 1.4 Assumptions and applicability

The doctrine consolidates current New Zealand practice from training, volunteer and Police operational experience, IMT tutors and national and international publications. It is deliberately not exhaustive.

It applies to all LandSAR IMT personnel and assumes endorsement by Police and RCCNZ. It rests on a **federation model**:

| Who | Role |
|---|---|
| National LandSAR | Strategy, standards, quality assurance, data systems, national partnerships |
| Local groups | Local identity and relationships, readiness, front-line response |
| Coordinating Authorities | Statutory control of the search and rescue operation (*SAROP*) |

## 1.5 1.5 Audience

The primary audience is volunteer IMT members. The doctrine also gives Police, RCCNZ and partner organisations a transparent view of how LandSAR manages operations, and other SAR organisations may use it as a reference.

## 1.6 1.6 Strategic context

New Zealand's terrain, weather, rivers, caves, alpine areas and visitor mobility demand local knowledge within a nationally coherent system. Coordination is split by jurisdiction:

| Category | Coordinated by | Typical operations |
|---|---|---|
| I | New Zealand Police (locally) | Land, subterranean, river, lake and close-to-shore |
| II | RCCNZ (nationally) | Missing aircraft, distress beacons, offshore marine |

The doctrine integrates established international search theory with local protocols and training. Joint Police, RCCNZ and LandSAR training is the key to a consistent approach.

The **keep it simple** principle applies throughout. Groups with high operational tempo can go further with extended search planning and advanced theory, provided their insights come back into the national doctrine.

*Condensed from draft v0.1, Part 1.*

CHAPTER

TWO

# PART 2 PRINCIPLES

The "why" behind what we do. These 15 principles guide decisions, support inter-agency work and drive continuous improvement. They sit at the top of the derivation hierarchy: every concept traces back to at least one of them.

Numbering follows the draft. The order and grouping of principles is still open for review.

## 2.1 PRN-01 Unified and consistent approach

**PRN-01 · Principle · version 1**

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Every team, wherever it is, follows the same recognised practices and processes. Consistency makes large operations work, lets groups and Coordinating Authorities operate together, and gives a clear basis for national training, competencies and operational reviews.

Local groups bring resources and knowledge of terrain, community and weather. National doctrine, training, systems and assurance make that knowledge interoperable and scalable.

**In short:** set the principles once, apply them consistently everywhere.

*Condensed from draft v0.1, Part 2, principle 1.*

## 2.2 PRN-02 Adherence to a national management framework (CIMS)

**PRN-02 · Principle · version 1**

- **Supports:** *CON-04*, *CON-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR incident management aligns with the *CIMS*, applied flexibly to the land SAR environment. Terms and acronyms follow the CIMS manual, with LandSAR-specific terms only where needed.

CIMS is a scalable management structure. It does not say what actions to take, what the priorities are, or how to reason about the subject, environment and resources. That comes from SAR knowledge and experience.

**In short:** CIMS is needed for multi-agency interoperability, but it is not a SAR manual.

*Condensed from draft v0.1, Part 2, principle 2.*

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## 2.3 PRN-03 Fostering partnerships and relationships

**PRN-03 · Principle · version 1**

- **Supports:** *CON-05*, *CON-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR is a **support agency** to Police and *RCCNZ* under a memorandum of understanding. The Coordinating Authorities hold authority and legislative responsibility; LandSAR provides operational expertise, tactical advice, specialised people (including trained IMTs) and local knowledge. LandSAR also supports other agencies where appropriate, for example in civil defence emergencies.

Diverse IMTs, with varied backgrounds and perspectives, avoid groupthink and produce more robust strategies.

**In short:** what our partnerships look like to us.

*Condensed from draft v0.1, Part 2, principle 3.*

## 2.4 PRN-04 Importance of regional collaboration

**PRN-04 · Principle · version 1**

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Groups meet proactively, share expertise and stay aligned with national standards, so they can call on each other quickly during operations. No team is isolated; all benefit from the collective experience of IMTs nationally.

**In short:** none of us is smarter than all of us.

*Condensed from draft v0.1, Part 2, principle 4.*

## 2.5 PRN-05 Proactive planning and readiness

**PRN-05 · Principle · version 1**

- **Supports:** *CON-03*, *CON-13*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

IMTs pre-plan for likely events in predictable or repeat locations, and practise those plans. Every operation should have a suitably trained and resourced IMT. Never assume an operation will be over quickly: set up the team and processes as if it will be protracted.

**In short:** pre-plan to improve efficiency, and resource the IMT properly.

*Condensed from draft v0.1, Part 2, principle 5.*

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## 2.6 PRN-06 Effective and supportive leadership

**PRN-06 · Principle · version 1**

- **Supports:** *CON-05*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Leadership matters inside the team (cohesion and clarity) and between teams (trust and coordination with other groups and agencies). Good leaders collaborate, communicate clearly, keep situational awareness and seek expertise when needed. They empower team members to contribute.

**In short:** leadership affects operational outcomes; develop good leaders inside and outside the organisation.

*Condensed from draft v0.1, Part 2, principle 6.*

## 2.7 PRN-07 The critical role of communications

**PRN-07 · Principle · version 1**

- **Supports:** *CON-10*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The overriding communications principle is **information superiority**: one reliable, integrated flow of intelligence to the IMT, giving shared situational awareness and enabling safe, coordinated deployment by all agencies.

**In short:** without intelligence coming in, the IMT cannot predict, plan, resource or oversee effectively.

*Condensed from draft v0.1, Part 2, principle 7.*

## 2.8 PRN-08 Evidence-based and intelligence-led operations

**PRN-08 · Principle · version 1**

- **Supports:** *CON-08*, *CON-09*, *CON-11*, *CON-17*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations rest on evidence and are guided by intelligence. Analytical methods give structure, complemented by the knowledge and intuition of experienced people. The same evidence-based approach applies to building doctrine, training, planning and deployment.

**In short:** use evidence and intelligence to support or refute our thinking and actions.

*Condensed from draft v0.1, Part 2, principle 8.*

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## 2.9 PRN-09 Search urgency

**PRN-09 · Principle · version 1**

- **Supports:** *CON-04*, *CON-15*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The *decision to act* is urgent, even if the first action is only gathering information. Urgency never overrides safety or methodical planning: initial objectives avoid unnecessary risk and unproductive tasking.

The Coordinating Authority usually sets initial urgency shortly after notification. The IMT can review and formalise it during the initial response.

**In short:** decide urgently, then plan a response that fits the circumstances.

*Condensed from draft v0.1, Part 2, principle 9.*

## 2.10 PRN-10 Structured information management

**PRN-10 · Principle · version 1**

- **Supports:** *CON-11*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Built on the *COP*, information management rests on three tests:

- **Access**: everyone who needs information can see it.
- **Accuracy**: intelligence is grounded in verified evidence; displaying it visually helps catch errors.
- **Actionability**: information is analysed and presented clearly enough to guide objectives.

**In short:** I need to see what you see. Record what you are doing and why.

*Condensed from draft v0.1, Part 2, principle 10.*

## 2.11 PRN-11 Systematic and methodical approach: POA × POD = POS

**PRN-11 · Principle · version 1**

- **Supports:** *CON-07*, *CON-08*, *CON-12*, *CON-13*, *CON-14*, *CON-15*, *CON-16*, *CON-17*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations aim to maximise *POS*. In the initial response this generally follows the objectives of the Bike Wheel model: actions that statistically give the best chance of success. If the subject is not found, the operation moves to extended search planning, where areas with high *POA* are searched in priority order with resources that give the highest *POD*.

Focus on what can be controlled: intelligence gathering, a detailed subject profile, local knowledge and lost person behaviour data, and plausible scenarios that direct the search. Record every plan, action and decision. Proven processes support, but never replace, the IMT's judgement.

**In short:** focus on what you can control, trust the process, and work in small, well-considered steps.

*Condensed from draft v0.1, Part 2, principle 11.*

## 2.12 PRN-12 The principle of simplicity

**PRN-12 · Principle · version 1**

- **Supports:** *CON-15*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Processes must be simple to understand and easy to apply in real time. Operational tempo and experience vary widely across the country, so any complication should be refined into a simple process that keeps the underlying concept.

**In short:** keep it simple. It removes ambiguity and builds confidence.

*Condensed from draft v0.1, Part 2, principle 12.*

## 2.13 PRN-13 Prioritising safety

**PRN-13 · Principle · version 2**

- **Supports:** *CON-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Some risk may be accepted to save a saveable life, but the safety of LandSAR personnel is non-negotiable. A robust safety culture includes dynamic risk assessment, fatigue management, thorough briefings, communications discipline and medical care for teams and the subject. Volunteers are only deployed on tasks they are trained and qualified for.

“Our people get to come home.” “We are in the business of saving lives, not trading them.”

**In short:** safety is an IMT obligation, not a word. It’s their emergency, but our job.

### 2.13.1 Version history

- **v2** (illustrative): added fatigue management and communications discipline. *CON-02 Risk management and safety* was written against v1, so it is listed as needing review.
- **v1**: first version.

*Condensed from draft v0.1, Part 2, principle 13.*

## 2.14 PRN-14 Continuous improvement

**PRN-14 · Principle · version 1**

- **Supports:** *CON-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The doctrine runs on a feedback loop. Every operation and *SAREX* is debriefed for lessons and actionable insights. These improve local performance, then go to a national subject-matter-expert review point, and on to Police and LandSAR IMT leads for ratification and a doctrine update.

The loop is driven by the sector stakeholders who live with the operational consequences.

**In short:** improve nationally, and let others learn from our mistakes and achievements.

*Condensed from draft v0.1, Part 2, principle 14.*

## 2.15 PRN-15 Iwi and community partnerships

**PRN-15 · Principle · version 1**

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Operations respond to community needs and include iwi and Māori involvement where whenua, whānau, tikanga, local knowledge or consequences require it.

**Note**

The draft has no principle statement for this item yet, and no concept derives from it, so the *Traceability status* page shows it as a coverage gap.

*Condensed from draft v0.1, Part 2, principle 15.*

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CHAPTER

THREE

# PART 3 CORE CONCEPTS AND FRAMEWORK

The "how" of land search management, from fundamental concepts down to guidelines, processes, roles and capability.

## 3.1 3.1 Fundamental concepts

The central ideas, theories and approaches at the core of the doctrine. Numbering follows the draft. The 3.1 order, and how content is split between 3.1 and 3.2, are still under review.

### 3.1.1 CON-01 Hierarchical framework of the doctrine

**CON-01 · Concept · version 1**

- **Derives from:** *PRN-01* v1; *PRN-04* v1; *PRN-14* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The doctrine cascades from enduring principles to specific operational functions:

| Level | What it is |
|---|---|
| Principles | Foundational beliefs and intent behind the approach |
| Fundamental and core operational concepts | The central theories; the "how" |
| Guidelines | Recommended practice for a typical operation; not mandatory |
| Processes and procedures | Established sequences of action and SOPs |
| Systems | The frameworks that enable processes |
| Functions | Individual tasks within processes and systems |

The whole runs as a **double-loop learning system**: lessons found at the functional level flow upward and can change processes, guidelines, concepts and even principles. This site implements that loop. See *How this doctrine is maintained*.

#### Proposed national IMT framework

A national framework would professionalise IMTs, give a career pathway and support succession:

- **IMT development pathway**: training and experience required at each level of responsibility.
- **National tutor pool**: tutors drawn from experienced, current IMT practitioners.
- **Regional representatives**: the subject-matter-expert collaboration point for consistent practice.
- **National IMT coordinator**: gives every group a voice and turns operational lessons into doctrine changes.

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**Note**

The level names in the draft text, its figure and the doctrine brief don't yet match. This site uses principle → concept → guideline → process → form.

*Condensed from draft v0.1, Part 3.1 §1.*

### 3.1.2 CON-02 Risk management and safety

**CON-02 · Concept · version 1**

- **Derives from:** *PRN-13* v1 (**review needed:** PRN-13 is now v2)
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Safety is governed by the LandSAR Safety Management System (*SMS*), which covers training, exercises and operations, and meets both health and safety law and the memorandum of understanding with the Coordinating Authorities. Following it is mandatory for every member.

Safety management starts at the beginning of every operation and continues until everyone is back to normal business. Briefings happen at three levels: the incident controller or safety officer (strategic hazards and reporting), operations (task-specific hazards) and team leaders (how the team will manage risk).

#### IMT obligations

1. Cover safety in every team briefing.
2. Assess and record operation-level hazards.
3. Prepare a safety plan, and use it to populate the hazard section of each tasking.
4. Monitor progress, update teams on changing hazards and look after their wellbeing.
5. Debrief teams on safety issues.
6. Record and report accidents and incidents.
7. Stay current with safety advisories and the LandSAR safety standards.

Teams manage dynamic risk in the field with the Take 5 process. Groups benefit from an annual hazard register for their area, so briefings can focus on task-specific risks.

| Risk domain | Example controls |
|---|---|
| Operational environment | Weather and river thresholds, daylight limits, go/no-go decisions |
| Fatigue | Shift planning, rest periods, replacement teams, IMT fatigue monitoring |
| Comms and tracking | Scheduled calls, check-in/out, team tracking, lost team procedure |
| Psychological wellbeing | Pre-briefs for traumatic exposure, peer and post-incident support |
| Legal and ethical | Privacy, media discipline, scene preservation, safeguarding |
| Volunteer sustainability | Reasonable call-out expectations, employer support, avoiding burnout |

*Condensed from draft v0.1, Part 3.1 §2.*

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### 3.1.3 CON-03 Pre-operational readiness plans

**CON-03 · Concept · version 1**

- **Derives from:** *PRN-05* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

IMTs develop and practise pre-plans for predictable events and locations, based on the area's SAR history. Pre-plans reduce the research and planning burden when an urgent response is needed. They can also cover how the IMT will manage civil defence responses or large numbers of spontaneous volunteers.

A pre-plan typically covers:

- activation and mobilisation
- the area, its geography, access points and typical activities
- initial actions and resources for common scenarios
- health and safety, and environmental conditions
- previous searches in the area
- communications plan and coverage
- key contacts, facilities, medical and logistics resources.

Keep pre-plans brief, and structure them consistently so people can find information quickly.

*Condensed from draft v0.1, Part 3.1 §3.*

### 3.1.4 CON-04 Notification and awareness

**CON-04 · Concept · version 1**

- **Derives from:** *PRN-02* v1; *PRN-09* v1
- **Supports:** *PRC-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Notification starts every operation. Reports usually reach a Coordinating Authority directly or through partners such as ambulance services or the Department of Conservation. All reports should be channelled to the Coordinating Authority, so urgency and response can be assessed with local knowledge.

Whoever receives and assesses the report becomes the **incident controller** and holds all *CIMS* functions until they delegate them. Their first jobs are to record the key information (the Missing Person Report, *MPR*, is a good guide) and to assess the likely scale and urgency.

- **Urgent and obvious**: deploy immediately using local knowledge and available assets.
- **Not obvious**: work through the first steps of the *initial response process* (MPR, map, information collection, obvious scenarios) before finalising search urgency, because urgency scored on thin information tends to overstate it.

Avoid planning alone. A LandSAR IMT can be engaged early to support these decisions, and should always set up as if the operation will be protracted.

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**Note**

Whether notification is in scope for this doctrine is still to be confirmed against the doctrine brief.

*Condensed from draft v0.1, Part 3.1 §4.*

### 3.1.5 CON-05 Leadership

**CON-05 · Concept · version 1**

- **Derives from:** *PRN-06* v1; *PRN-03* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

LandSAR works as a support agency to the Coordinating Authorities, who hold ultimate authority. Within that relationship, leadership is a cornerstone of a high-performing IMT, both within the team and between teams and agencies.

Strong leaders collaborate, communicate clearly, maintain situational awareness and seek expertise when needed. Leadership is developed through training, experience on the job and succession planning.

*Condensed from draft v0.1, Part 3.1 §5.*

### 3.1.6 CON-06 Incident management teams and CIMS

**CON-06 · Concept · version 1**

- **Derives from:** *PRN-02* v1; *PRN-03* v1
- **Supports:** *GDL-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

An IMT plans and manages the whole operation so it is safe, efficient and effective. An IMT should be set up for every operation, whatever its scale, so no one plans alone. Structure and formality scale with the operation: be practical at the start, then formalise roles as complexity and span of control grow.

| CIMS function | Role in a land search |
|---|---|
| Control | Overall strategy, objectives, safety and coordination; appointed by the lead agency. A response manager can run the IMT day to day. |
| Intelligence | What we know, what we need to know; subject profile, planning points, weather, terrain, field intelligence; builds the *COP* |
| Planning | Where to search next; *IAP*s, long-term and contingency plans |
| Operations | Tasking, coordinating and supervising field teams and air assets |
| Logistics | People, equipment, transport, food, facilities; responder wellbeing before and after tasks |
| Safety | Hazard identification and a safe work culture across all agencies |
| Welfare | Support for the subject's family and the affected community |
| Public information | Media and community messaging, usually with Police |

#### Getting search teams to the right place

- Know the capability of local and neighbouring resources, and engage them early.
- Seek expert advice for unfamiliar tasks or resources.
- Review each operational period to check intelligence has been acted on.
- Use limited resources early to gather information.
- Use trained IMT members, and assume a protracted operation from the start.
- Work with local communities.

Diverse teams, with field, management and analytical backgrounds, make better decisions and avoid groupthink.

*Condensed from draft v0.1, Part 3.1 §6.*

### 3.1.7 CON-07 Phases of a land search and rescue operation

**CON-07 · Concept · version 1**

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-04*, *GDL-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

An operation is a continuum, but each phase needs different thinking:

1. **Initial response (*IR*)**: pre-determined actions based on lost person behaviour and the *Bike Wheel model*; CIMS functions applied informally and scaled to need.
2. **Extended search planning (*ESP*)**: starts if the initial response doesn't find the subject. Planning shifts from the statistical subject to *this* subject, using scenarios, consolidated intelligence and methodical searching of probable areas.
3. **Rescue (known location)**: can start at any point when the subject is found, or as a stand-alone operation, for example a beacon activation.

Notification isn't a phase as such; it usually sits with a Coordinating Authority.

Intelligence is the fuel. Without enough focus on collecting and analysing it, a search gets stuck in the initial response, tasking resources reflexively with no evidence to prefer one place over another. Move to extended search planning as the initial response objectives are completed, and resource intelligence from the start so the transition is smooth.

**Note**

The draft uses several names for the second phase and differs on whether "operational conclusion" is a fourth phase. This site uses *extended search planning* throughout.

*Condensed from draft v0.1, Part 3.1 §7.*

### 3.1.8 CON-08 Maths in SAR

**CON-08 · Concept · version 1**

- **Derives from:** *PRN-08* v1; *PRN-11* v1

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Numbers run through land SAR: search area and travel distance, search urgency scores, lost person behaviour statistics, scenario weighting and probability of area. People interpret probability *words* very differently; numbers remove that ambiguity, even when the judgement behind them is subjective.

*POD* is often reported as High, Medium or Low. Words alone can't be used to recalculate *POA* after a segment is searched, which is central to progressive search management. The doctrine therefore binds each word to a number. A simple prompt: "If there were 10 people in your area, how many would you have seen?" Eight means 80 %.

| Rating | POD as drafted |
|---|---|
| High | 66–99 % |
| Medium | 33–66 % |
| Low | 1–33 % |

*This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.*

Recalculating probability is covered in *CON-17 Search theory: POA × POD = POS*.

*Condensed from draft v0.1, Part 3.1 §8.*

### 3.1.9 CON-09 Lost and missing person behaviour

**CON-09 · Concept · version 1**

- **Derives from:** *PRN-08* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Lost person behaviour (*LPB*) research, built from large international incident databases, moves initial planning from instinct to data. It lets an IMT set objectives and send resources to the most likely places first, when little is known about the subject. It is a tool, not the answer.

Apply it with judgement:

- The data is aggregated. Specific knowledge of this subject (fitness, injury, state of mind, intentions) can outweigh it.
- Local SAR history matters: subjects cluster where terrain and common decision errors take them.
- Check the **ecoregion** and the **sample size** (n) behind each figure; small samples are less reliable.
- Choose the **subject category** (for example hiker, child, despondent, dementia) using the category hierarchy; several may need considering.
- Use the category's **investigative questions** and **initial tasks**.
- Apply the category tables to the search area: distance travelled, elevation change, mobility, dispersion, find location, scenarios, track offset and survivability.

LandSAR runs dedicated training on applying LPB.

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**Note**

Figures for New Zealand's share of the data, and success rates attributed to applying the model, are being verified and are not reproduced here.

*Condensed from draft v0.1, Part 3.1 §9.*

### 3.1.10 CON-10 Communications

**CON-10 · Concept · version 1**

- **Derives from:** *PRN-07* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Communications connect every level, from the subject to the incident controller. The IMT is the hub: it receives distress signals, status reports and clues, turns them into intelligence, coordinates internally, and sends a clear action plan back out. That feedback loop validates the strategy and lets it adjust.

The IMT needs clear, concise:

- briefings in a standard format, with safety included
- taskings with scheduled reporting times
- status reports and structured IMT meetings
- handovers to incoming IMTs
- radio or satellite communication with field teams
- planned debriefs of every team.

Field teams provide radio checks, situation reports on schedule, early intelligence on clues and conditions, and a POD from each searcher after the task.

| Capability | Doctrine position |
|---|---|
| Radio | Interoperable channels, disciplined procedure, fallbacks for terrain shadow |
| Mapping and GIS | One common picture of tasks, tracks, clues, hazards and probability areas |
| Drones and aircraft | Use where lawful, safe and relevant; don't overclaim effectiveness in dense bush |
| Mobile and digital data | Use Police-approved processes; treat digital clues as perishable and private |
| Data protection | Role-based access, minimal disclosure, secure storage |

Prefer simple, robust systems that work when infrastructure is degraded.

*Condensed from draft v0.1, Part 3.1 §10.*

### 3.1.11 CON-11 Evidence-based and intelligence-led operations

**CON-11 · Concept · version 1**

- **Derives from:** *PRN-08* v1; *PRN-10* v1
- **Supports:** *GDL-01*

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

- **Evidence-based**: choose strategies by appraising proven data (historical SAR statistics, research, professional expertise) rather than habit.
- **Intelligence-led**: continuously collect, process and analyse information during the operation, and use it to set priorities, adjust resources and change tactics.

Early in the initial response, statistics drive most actions. As the operation progresses, intelligence makes the strategy specific to this subject and this terrain. Record every plan, action and decision throughout.

The intelligence function turns information into intelligence under three tests: access, accuracy and actionability. Source reliability can be graded where people are trained to do so.

#### Common operating picture

The *COP* is the single, shared display of the incident: everyone works from the same verified facts, decisions are faster, and the IMT can see coverage and team positions at a glance. A SAR COP typically shows:

1. timeline (weather, subject actions, SAR actions)
2. current incident action plan
3. subject profile
4. information collection plan
5. weather and terrain overlays
6. planning map (planning points, hazards, attractants, clues, regions and segments, statistical rings, search effort)
7. operations map (team positions and status)
8. planning data (urgency, LPB category, scenarios, clues)
9. logistics and resource tracking.

*Condensed from draft v0.1, Part 3.1 §11.*

### 3.1.12 CON-12 Management by objectives

**CON-12 · Concept · version 1**

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-01*, *GDL-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Management by objectives (*MBO*) keeps every action tied to the mission, through short, time-bound operational periods. It is consistent with CIMS. Teams that jump straight to tasks usually have objectives in mind; writing them down creates clarity.

Remember ***MOST***:

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| | Question |
|---|---|
| **M**ission | Why: for example “locate the missing person” |
| **O**bjectives | What will we do this period? |
| **S**trategies | How will we do it? |
| **T**asks | Who will do it? |

Objectives should be **SMART**: specific, measurable, achievable, relevant and time-bound.

Short operational periods (often 8–12 hours, aligned with Coordinating Authority shifts) allow the plan to adapt to new information, manage fatigue, make progress measurable, and bring fresh thinking as IMTs rotate.

*Condensed from draft v0.1, Part 3.1 §12.*

### 3.1.13 CON-13 Incident action plans

**CON-13 · Concept · version 1**

- **Derives from:** *PRN-11* v1; *PRN-05* v1
- **Supports:** *GDL-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The incident action plan (*IAP*) puts MBO into practice: a concise, forward-looking plan for one operational period that balances probability, detection, survivability, hazards, urgency and the consequences of being wrong. The planning function leads it, with input from every function.

- **Urgent searches**: deploy first and back-capture the IAP. Uncertainty is not a reason for delay.
- **Protracted searches**: write the IAP before teams are briefed and deployed.
- **Handovers**: never hand over without an IAP for the next period.

#### Building an IAP

1. **Set the operational period first**, allowing for travel and debrief. For example, a 10-hour period might be 2 hours travel, 4–6 hours searching, 2 hours return and debrief. Searching much longer than 4–6 hours degrades POD.
2. **Set prioritised SMART objectives**, typically covering search, safety, management and investigation.
3. **Develop the plan**: objectives (what), strategies (how), tasks (who).
4. **Task only to objectives.** If a new task emerges, change the plan formally rather than tasking ad hoc.
5. **Debrief and feed back** into the next IAP.

Use a simple one-page IAP per operational period, filed with that period’s tasks and debriefs, so the record builds period by period.

*Condensed from draft v0.1, Part 3.1 §13.*

### 3.1.14 CON-14 The Bike Wheel model

**CON-14 · Concept · version 1**

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The Bike Wheel is a visual model that turns lost person behaviour statistics into initial response objectives. It comes from international search theory and has been adapted in New Zealand to include management and intelligence objectives.

| Wheel element | Initial response objective |
|---|---|
| Hub | Process the planning point and search close in |
| Spokes | Search the intended route and travel corridors |
| Reflectors | Check hazards, attractants and areas with SAR history |
| Rim, with rocks at corridor ends | Contain the area at natural containment points |
| Stick in the wheel | Search areas suggested by early scenarios |
| Frame | Build the profile, establish comms, prepare for handover |

#### Example initial response objectives (one operational period)

1. Complete the information collection plan and build the subject profile.
2. Establish reliable communications for the search area.
3. Contain the search area.
4. Process the planning point and complete a close-in search.
5. Search the area suggested by the initial scenario.
6. Search the intended route and check decision points.
7. Search travel corridors next to the intended route.
8. Check known hazards, attractants and SAR history locations.
9. Prepare the IAP for handover.

In extended search planning the same wheel is reused: what is left is “the spaces between the spokes”, which are reduced, divided into regions and segments, and weighted.

**Note**

Fixed distances shown in some versions of the figure, and attribution, are still being checked.

*Condensed from draft v0.1, Part 3.1 §14.*

### 3.1.15 CON-15 The initial response process

**CON-15 · Concept · version 1**

- **Derives from:** *PRN-09* v1; *PRN-11* v1; *PRN-12* v1
- **Supports:** *GDL-02*, *PRC-01*

- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

The initial response begins straight after notification and follows pre-determined actions guided by the *Bike Wheel*. Its length depends on the size and complexity of the area, not a fixed time.

IMT managers need a process they can repeat without reference material. The mnemonic is ***MMISSI TD***:

1. **M**issing person report: initial details, medical, equipment, intentions.
2. **M**ap: terrain analysis, hazards, LPB search area, travel time and distance, planning points.
3. **I**nformation collection plan: build the profile, fill gaps, assign enquiries.
4. **S**cenario: quick, obvious scenarios.
5. **S**earch urgency: confirm the initial assessment still holds.
6. **I**ncident action plan: set objectives with the Bike Wheel.
7. **T**ask: from the objectives; preserve the planning point and clues; contain.
8. **D**ebrief: gather intelligence and POD.

Repeat steps 6–8 until the initial response objectives are met. The step-by-step version is *PRC-01 Initial response (MMISSI TD)*.

**Note**

How urgent cases bypass the sequence, and how steps run in parallel, is still being worked through.

*Condensed from draft v0.1, Part 3.1 §15.*

### 3.1.16 CON-16 The extended search planning process

**CON-16 · Concept · version 1**

- **Derives from:** *PRN-11* v1
- **Supports:** *GDL-05*, *PRC-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Extended search planning builds a progressive plan on the results of earlier search effort. In outline:

1. Review the initial response (start while it winds down).
2. Determine the search area by subjective and deductive reasoning.
3. Develop scenarios that fit the evidence, LPB data and profile.
4. Mark scenario and geographical regions on the map.
5. Weight the regions.
6. Divide the highest-weighted regions into segments.
7. Weight the segments.
8. Task to the highest-weighted segments.
9. Debrief teams to establish POD.

10. Recalculate segment weights from POD and new intelligence.
11. Re-task to the next highest regions or segments.

Repeat steps 8–11. Because every segment carries a number, segments in different regions can be compared directly. The step-by-step version is *PRC-02 Extended search planning cycle*.

*Condensed from draft v0.1, Part 3.1 §16.*

### 3.1.17 CON-17 Search theory: POA × POD = POS

**CON-17 · Concept · version 1**

- **Derives from:** *PRN-08* v1; *PRN-11* v1
- **Supports:** *GDL-05*, *GDL-06*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Search planning comes down to two things: finding the right place to search, and applying resources that can detect what you are looking for, at the right time.

$$POS = POA \times POD$$

- *POA*: the likelihood the subject is in this region or segment.
- *POD*: the likelihood this resource would find the subject if they were there.
- *POS*: the overall likelihood of success.

#### Closed search system

New Zealand uses a **closed search area**: the IMT defines a bounded area and assumes the subject is inside it. The rest of the world is not forgotten; it is covered by investigation and public appeals, not by searching.

#### Search area, regions and segments

In the initial response the search area is set by LPB statistical rings, theoretical travel distance and any early scenarios. In extended search planning it is refined into a shape that follows terrain and evidence; the rings remain for reference only.

The refined area is divided into about 6–10 **regions** along clear terrain features: scenario-based regions (usually higher POA) and geographical regions. High-POA regions are divided into **segments**, each sized for a 4–6 hour task.

#### Scenarios and weighting

A scenario is a story that runs from the planning point to a specific piece of terrain. “Lost somewhere in the area” is not useful. Present all the evidence to the IMT first, then test each scenario: does it fit the facts, is it realistic, is it consistent with LPB and the profile, is it achievable in this terrain, and what evidence contradicts it?

Weight scenarios with the *scenario weighting sheet* or, where evidence is complex, with analysis of competing hypotheses (*ACH*), which looks for the scenario *least contradicted* by the evidence. The two are complementary.

#### Recalculating after a search

When a segment is searched and the subject isn’t found, its POA drops:

$$POA_{new} = POA_{old} \times (1 - POD)$$

For example, a segment weighted 14 searched at 80 % POD drops to 14 × 0.2 = 2.8. A second search with a different resource reduces it further. The IMT, not the field team, owns the final POD: it checks the debrief against GPS tracks, photos and conditions before using it, because an inflated POD moves the search away from a poorly searched area.

| Rating | POD as drafted |
|---|---|
| High | 66–99 % |
| Medium | 33–66 % |
| Low | 1–33 % |

*This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.*

**Note**

How ACH results translate into segment weights, and tools for more objective POD, are open questions in the draft.

*Condensed from draft v0.1, Part 3.1 §17.*

## 3.2 3.2 Core operational concepts

The concepts from 3.1, applied in an operational context. The draft lists the topics; content is still to be written.

**Not yet drafted**

Each topic will become an item page that derives from the 3.1 concept it applies.

1. IMT set-up and the common operating picture
2. CIMS in practice
3. Lost person behaviour
4. Information management and intelligence
5. Search urgency
6. Planning and maps
7. Planning points
8. Scenario development
9. Search area determination in IR and ESP
10. Close-in search
11. Containment
12. Decision points
13. Terrain analysis
14. IAPs and the Bike Wheel
15. Search methods
16. Tasks, briefing and the mobility matrix

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17. Search POD and coverage (search effort evaluation)
18. Rescue
19. Team debriefing
20. Handovers
21. Conclusion: demobilisation, suspension, welfare, actionable insights and operational debriefs

## 3.3 3.3 Guidelines

Guidelines are a consolidated overview of the recommended process for a typical land search. They are not mandatory, but they support consistency and effective practice. Pick the section relevant to you on the day.

Guidance tier: one approval from any reviewer merges a change.

### 3.3.1 GDL-01 Managing an IMT: key actions

**GDL-01 · Guideline · version 1**

- **Derives from:** *CON-06* v1; *CON-11* v1; *CON-12* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Carry these out continuously, from notification to conclusion, in every operation:

- **Develop the subject profile** through ongoing interviews and investigation.
- **Manage risk dynamically** until everyone is home after the final debrief.
- **Scale the structure**: apply *CIMS* in proportion to the operation.
- **Prioritise the four Cs** in the initial response: confirm the planning point, close-in search, containment, communications. Immediate hazards or a strong scenario can take priority.
- **Document everything**: intelligence, decisions, resources, actions, debriefs.
- **Keep a common operating picture** on walls or shared displays, not hidden on individual screens.
- **Plan for a long operation** from the outset.
- **Engage help early**: catching up once behind is very hard.
- **Treat search teams as intelligence sources**: look for clues and sign, not just the subject.
- **Manage media and social channels** with a dedicated resource, usually through Police.
- **Use LPB data and the Bike Wheel** in the initial response.
- **Process before forms**: keep the method going even when forms or technology are unavailable.
- **Manage by objectives**: don't confuse being busy with being effective.

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#### Setting up

| Topic | Guideline |
|---|---|
| Notification | The Coordinating Authority is incident controller until it delegates. Use the MPR. If urgent, deploy now; otherwise put teams on standby and follow the initial response process. Avoid planning alone. |
| Safety | Develop a safety plan early. Appoint a safety officer if the environment is beyond the controller's expertise. Only send trained and equipped people. |
| Pre-plans | Check for and action any relevant pre-plan. |
| Incident control point | Choose a location with access, power, comms, display space and room to scale. Staff the IMT so no function is compromised. |
| Forward base | Decide its purpose first. Watch for isolated functions, loss of control by the IMT and intelligence that doesn't get passed on. |

*Condensed from draft v0.1, Part 3.3.1.*

### 3.3.2 GDL-02 Initial response phase

**GDL-02 · Guideline · version 1**

- **Derives from:** *CON-15* v1; *CON-14* v1
- **Supports:** *FRM-03*, *PRC-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Follows the *initial response process*. Start an operations log, decision log and timeline straight away, and display everything on the common operating picture.

| Step | Guideline |
|---|---|
| 1. Missing person report | Complete or update the MPR. Build the profile and choose an LPB category. |
| 2. Map | Mark planning points, close-in search area, LPB rings and intended route. Analyse travel corridors, hazards, weather effects on terrain and subject, barriers and attractants. |
| 3. Information collection plan | Address the gaps from steps 1 and 2. Assign enquiries, record results, display key intelligence. |
| 4. Initial scenarios | Quick and obvious stories that point to a piece of terrain: start point, direction of travel, route, activity, destination. Rank by consensus. |
| 5. Search urgency | Confirm or update the Coordinating Authority's assessment. Consider go/no-go for responder risk: don't make their emergency your emergency. |
| 6. Incident action plan | Set the operational period and mission, consider the factors, write SMART objectives, strategies and tasks. |
| 7. Tasking | Specific, unambiguous tasks of 4–6 hours. State whether speed, efficiency or thoroughness matters most (*SET*). Brief teams together in a standard format. |
| 8. Debrief | The most critical step. Debrief the whole team, not just the leader. Record POD from each searcher. Feed results into the next IAP. |

#### Recording POD at the debrief

Ask each searcher for a rating, then confirm it as a number. Average the individual values; outliers show where re-searching is needed.

| Rating | POD as drafted |
|---|---|
| High | 66–99 % |
| Medium | 33–66 % |
| Low | 1–33 % |

*This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.*

*Condensed from draft v0.1, Part 3.3.2.*

### 3.3.3 GDL-03 Handovers, demobilisation and operational debriefs

**GDL-03 · Guideline · version 1**

- **Derives from:** *CON-13* v1; *CON-12* v1
- **Supports:** *FRM-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

**Note**

This section is empty in the draft. The points below come from the draft's summary checklist.

#### Handover

- Prepare the briefing area and update maps.
- Complete the operational period summary.
- The incoming IMT arrives early; the outgoing IMT briefs it using the IAP structure.
- The incident controller stays on briefly after handover.

#### Demobilisation

- Set release priorities and plan safe extraction of teams.
- Keep communications until everyone is accounted for.
- Return equipment and complete sign-out.
- Look after welfare: food, warmth, rest, critical incident stress.
- Hold a hot debrief and schedule a formal debrief.

*Condensed from draft v0.1, Part 3.3.3.*

### 3.3.4 GDL-04 Rescue phase: known location

**GDL-04 · Guideline · version 1**

- **Derives from:** *CON-07* v1
- **Owner:** doctrine-group · **Last reviewed:** 2025-06-30

Rescue can start at any point, or as a stand-alone operation such as a beacon activation where the nature of the emergency is unknown until the party is reached. Finding the subject changes the mission but doesn't end the operation.

1. **Locate**: confirm position precisely, assess hazards, notify the IMT, protect the scene, prevent unnecessary convergence.
2. **Stabilise**: first aid, protection from the environment, reassurance, ongoing monitoring.
3. **Plan extraction**: walk-out, stretcher, rope, helicopter, boat or vehicle, matched to the subject's condition, weather, daylight, terrain and responder fatigue.
4. **Transfer care** with a structured medical handover, recording times, interventions and condition changes.
5. **Family and media** go through Police and the IMT.

If a fatality or crime is possible, preserve evidence, avoid unnecessary movement, record observations and follow Police direction.

*Condensed from draft v0.1, Part 3.3.4.*

### 3.3.5 GDL-05 Extended search planning phase

**GDL-05 · Guideline · version 1**

- **Derives from:** *CON-16* v1; *CON-17* v1
- **Supports:** *FRM-03*, *PRC-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Move to extended search planning as the initial response objectives are completed; start planning before they finish so the transition is seamless.

| Step | Guideline |
|---|---|
| Review the IR | Map areas searched, clues and POD. Check paperwork. Identify gaps in intelligence and coverage, and whether re-searching with other techniques would help. |
| Search area | Move from circles to shapes: bound the area by bluffs, rivers, ridges and corridors, using the subject's profile, capability, equipment and clues. |
| Scenarios | Present all intelligence to the IMT first. Test each scenario against the facts, terrain and LPB. Record each on a scenario recording sheet and map it as a region. |
| Regions | Aim for about 6–10, mixing scenario and geographical regions. |
| Weighting | Use the *scenario weighting sheet* (option A) or ACH (option B). Write the weights on the map. |
| Segments | Segment the top one to three regions; weight segments using the same scenario. |
| Task and debrief | Task the highest segments; debrief whole teams for POD. |
| Recalculate | Reduce segment weights by POD. Re-search with a different resource, direction or timing, or move on. |

You will never reach 100 %. Deciding when to move on is a judgement informed by the numbers and the quality of the POD.

*Condensed from draft v0.1, Part 3.3.5.*

### 3.3.6 GDL-06 Suspension

**GDL-06 · Guideline · version 1**

- **Derives from:** *CON-07* v1; *CON-17* v1
- **Supports:** *FRM-03*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

**Note**

This section is under review with the Coordinating Authority.

Suspension is a Coordinating Authority decision. LandSAR members provide input and advice but don't make it.

Options are a **limited continuous search** (flyers, media), **scaling down** to fewer assets in a smaller area, or **suspending** until new information comes to light.

An IMT preparing suspension evidence provides:

- a summary of all search actions
- a map of areas searched, coverage and gaps
- updated POA and POD analysis
- clues found and their relevance
- scenario analysis and weighting outcomes
- a specialist survivability assessment
- a risk assessment for continued searching
- a record of family liaison and the family's wishes.

The operation is then peer reviewed by someone not involved, the family is told, and approval is sought from the appropriate authority for the operation's category.

*Condensed from draft v0.1, Part 3.3.6.*

The summary checklist (3.3.7) is a form: *FRM-03 IMT summary checklist*.

## 3.4 3.4 Processes and procedures

The established sequences of action that put the guidelines into practice. Each process derives from the concept and guideline it implements, so a change to either flags the process for review.

A land search runs through four stages: initial response, extended search planning, rescue, and operational conclusion (demobilisation, handovers, debriefs and, if needed, suspension).

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### 3.4.1 PRC-01 Initial response (MMISSI TD)

**PRC-01 · Process · version 1**

- **Derives from:** *CON-15* v1; *GDL-02* v1
- **Supports:** *FRM-01*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

| Step | Do | Output |
|---|---|---|
| 1 | Complete or update the missing person report | MPR, LPB category |
| 2 | Map and terrain analysis | Planning map with planning points, rings, route, hazards |
| 3 | Information collection plan | Assigned enquiries, updated profile |
| 4 | Initial scenarios | Ranked scenarios tied to terrain |
| 5 | Confirm search urgency | Search urgency form, go/no-go |
| 6 | Incident action plan | IAP with Bike Wheel objectives |
| 7 | Task | Tasking sheets, briefing |
| 8 | Debrief | *Team debrief* with POD |

Repeat steps 6–8 until the initial response objectives are met, then start *PRC-02 Extended search planning cycle*.

*Condensed from draft v0.1, Parts 3.1 §15 and 3.4.*

### 3.4.2 PRC-02 Extended search planning cycle

**PRC-02 · Process · version 1**

- **Derives from:** *CON-16* v1; *GDL-05* v1
- **Supports:** *FRM-01*, *FRM-02*
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

| Step | Do | Output |
|---|---|---|
| 1 | Review the initial response | Map of effort, clues, gaps |
| 2 | Determine the search area | Refined search area boundary |
| 3 | Develop scenarios | Scenario recording sheets |
| 4 | Mark regions | 6–10 scenario and geographical regions |
| 5 | Weight regions | *Scenario weighting sheet* |
| 6 | Segment top regions | Segments sized for 4–6 hour tasks |
| 7 | Weight segments | Segment POA |
| 8 | Task highest segments | Tasking sheets |
| 9 | Debrief | *Team debrief* with POD |
| 10 | Recalculate | Updated segment POA |
| 11 | Re-task | Next IAP |

Repeat steps 8–11 until the high-POA segments are searched to the IMT's satisfaction.

*Condensed from draft v0.1, Parts 3.1 §16 and 3.4.*

### 3.4.3 PRC-03 Notification and on-scene coordination

**PRC-03 · Process · version 1**

- **Derives from:** *CON-04* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

1. Reports go to the Coordinating Authority, which becomes incident controller.
2. The controller assesses scale, urgency and likely resources, ideally with an IMT rather than alone.
3. The Coordinating Authority engages LandSAR for field teams, an IMT, or both.
4. On-scene coordination:
    - If RCCNZ engages LandSAR without Police, a LandSAR response manager is assigned as on-scene coordinator (*OSC*).
    - If Police are involved and a LandSAR IMT is set up, Police can take the OSC role.

**Note**

The on-scene coordinator arrangements are being verified with the Coordinating Authorities.

*Condensed from draft v0.1, Part 3.4.*

### 3.4.4 Standard operating procedures

National SOPs or checklists proposed in the draft:

- helicopter briefing
- minimum field team and safety equipment
- “actions on” contingencies.

Groups may add local SOPs, for example: standing up the IMT, forward base management, spontaneous volunteer management, out-of-area deployment, search dog and drone deployment, and night operations.

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## 3.5 3.5 Roles and responsibilities

Who does what in relation to the doctrine.

| Actor | Role | Responsibilities for the doctrine |
|---|---|---|
| NZSAR Council | National strategic governance of the SAR system | Align sector guidelines with the doctrine; feed sector lessons into it |
| New Zealand Police | Coordinating Authority for Category I operations, including most land SAR | Endorse the doctrine; deploy LandSAR often to build capability; contribute to processes and debriefs |
| RCCNZ | Coordinating Authority for Category II operations; directs a growing number of beacon-initiated land operations, usually through an on-scene coordinator | Endorse the doctrine; integrate land teams where needed |
| Police SAR squads | Local knowledge, subject information, joint training with groups | Contribute at all levels, especially processes and debriefs |
| LandSAR national office | Custodian of the doctrine; supports the federation of volunteer groups | Maintain the doctrine; align training, systems and readiness reporting |
| Local LandSAR groups | Volunteer response capability and local knowledge | Train in and apply the doctrine; maintain readiness to agreed standards |
| Iwi, whānau, community and land managers | Local knowledge, tikanga, access, family connection | Engage early and respectfully, especially where whenua, marae, wāhi tapu or private land are involved |

The doctrine is freely available to sector partners, and every organisation with relevant expertise is invited to contribute through the continuous improvement process.

**Note**

IMT function roles (control, intelligence, planning, operations, logistics, safety, welfare, public information) are currently described in *CON-06 Incident management teams and CIMS*. Where they should sit is still under review.

*Condensed from draft v0.1, Part 3.5.*

## 3.6 3.6 Capability and resources

The doctrine supports existing national doctrine such as *CIMS*. To deliver consistent practice, all sector partners need to agree on it as the baseline, then refine it over time.

Beyond process, the draft identifies four enablers that need sector-wide commitment:

1. consistent national call-out and engagement procedures
2. an interoperable mapping tool
3. an interoperable incident management system
4. communication protocols, processes and resources.

Together these support activation and a common operating picture at local, regional and national level, and should extend to working alongside civil defence.

*Condensed from draft v0.1, Part 3.6.*

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CHAPTER

# FOUR

# PLANNED PARTS

Later parts are not in this review round. Topics known so far are listed so overlaps with Parts 2–3 can be resolved before drafting.

| Planned section | Overlaps with |
|---|---|
| Part 4 Application and implementation | |
| 4.1 Planning and preparation | *CON-03 Pre-operational readiness plans* |
| 4.3 Leadership | *PRN-06 Effective and supportive leadership*, *CON-05 Leadership* |
| 4.4 Communication | *PRN-07 The critical role of communications*, *CON-10 Communications* |
| 4.5 Training | National IMT framework in *CON-01 Hierarchical framework of the doctrine* |
| Part 5 Review and evolution | |
| 5.2 Lessons learned | *PRN-14 Continuous improvement* |

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CHAPTER

FIVE

# FORMS

Forms derive from processes or guidelines, so a change upstream flags the forms that need checking.

## 5.1 FRM-01 Team debrief (search effort evaluation)

**FRM-01 · Form · version 1**

- **Derives from:** *PRC-01* v1; *PRC-02* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Debrief the whole team in person where possible: one person asks (operations), one records (intelligence). Don't let teams fill this in themselves.

| Field | Notes |
|---|---|
| Team, task and segment | |
| Time in and out | |
| Route and coverage | Annotated map and GPS track |
| Technique and spacing | As tasked, or what changed and why |
| Conditions | Vegetation, visibility, weather, terrain |
| Clues | What, where, when; photographed and preserved? |
| Hazards found | Pass to the safety function |
| POD per searcher | Rating and number; average them |
| What limited detection | Fatigue, distractions, radio issues, terrain |
| Recommendations | Re-search, other resources, new areas |

| Rating | POD as drafted |
|---|---|
| High | 66–99 % |
| Medium | 33–66 % |
| Low | 1–33 % |

*This scale is defined once, in `_snippets/pod-scale.md`, and included on every page that shows it. The draft's band boundaries overlap at 33 and 66; fixing them here fixes every page.*

*Fields drawn from draft v0.1, Parts 3.1 and 3.3.*

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## 5.2 FRM-02 Scenario weighting sheet

**FRM-02 · Form · version 1**

- **Derives from:** *PRC-02* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

Use after the evidence has been presented to the whole IMT.

1. Each person independently gives their most likely scenario a score of 100.
2. They score every other scenario from 10 to 100, relative to it: 50 means half as likely. More than one scenario can score 100.
3. Add each scenario's scores.
4. Divide each total by the sum of all totals and multiply by 100 to get the weighting.

### 5.2.1 Worked example

| Scenario | Member 1 | Member 2 | Member 3 | Total | Weighting |
|---|---|---|---|---|---|
| A Injured near main track | 100 | 80 | 100 | 280 | 40 |
| B Side trip to a lake | 80 | 100 | 70 | 250 | 36 |
| C Lost in bush off track | 30 | 40 | 20 | 90 | 13 |
| D Sheltering off track | 20 | 30 | 30 | 80 | 11 |
| **Sum** | | | | **700** | **100** |

The weightings are relative values for setting priorities, written on the map as each region's POA.

*Method from draft v0.1, Part 3.1 §17. Example values are invented.*

## 5.3 FRM-03 IMT summary checklist

**FRM-03 · Form · version 1**

- **Derives from:** *GDL-02* v1; *GDL-03* v1; *GDL-05* v1; *GDL-06* v1
- **Owner:** doctrine-group · **Last reviewed:** 2026-10-01

A rapid-reference record for people who know the process.

### 5.3.1 Notification and safety

- Informant details recorded; MPR started
- Collaborating early, not planning alone
- Response decision made; teams on standby if not urgent
- Operations log, decision log and timeline started
- Safety plan started; safety officer appointed if needed

### 5.3.2 Incident control point and information

- Location chosen (power, comms, space, scalable)
- Common operating picture set up
- IMT roles assigned, including family liaison and media
- Profile, information collection plan and enquiries under way

### 5.3.3 Map and initial scenarios

- Planning points, close-in search area and route plotted
- LPB rings and theoretical distance plotted
- Hazards, attractants, barriers and corridors marked
- Quick scenarios ranked; most plausible set as an objective

### 5.3.4 Urgency, IAP and tasking

- Search urgency confirmed; go/no-go for responder risk
- Operational period and mission set; SMART objectives, strategies and tasks
- Tasks of 4–6 hours with map, subject summary and hazards
- Teams briefed together

### 5.3.5 Debrief

- Whole team debriefed; GPS, photos and maps reviewed
- POD recorded per searcher; clues analysed
- Results fed into the next IAP; POA updated

### 5.3.6 Extended search planning

- IR reviewed; search area refined
- Regions and segments defined and weighted
- Intelligence gathering continues

### 5.3.7 Handover, demobilisation and suspension

- Operational period summary complete; incoming IMT briefed
- Teams extracted and accounted for; welfare managed
- Hot debrief held; formal debrief scheduled
- If recommending suspension: evidence pack prepared (see *GDL-06 Suspension*)

*Condensed from draft v0.1, Part 3.3.7.*

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CHAPTER

SIX

# GLOSSARY

Each term is defined once here, and pages link to it with `{term}`POD``. Terms follow *CIMS* usage where CIMS defines them.

**ACH**
: Analysis of Competing Hypotheses. A structured method for testing scenarios against evidence, favouring the scenario least contradicted by it.

**CIMS**
: Coordinated Incident Management System. New Zealand's common framework for managing incidents across agencies.

**CIS**
: Close-In Search. A search of the immediate area around the initial planning point for the subject, clues and direction of travel.

**COP**
: Common Operating Picture. The single shared display of the incident: plans, maps, intelligence, resources and timelines.

**DOT**
: Direction of Travel.

**ESP**
: Extended Search Planning. The phase that follows the initial response when the subject has not been found.

**FLO**
: Family Liaison Officer.

**IAP**
: Incident Action Plan. The plan for one operational period: mission, objectives, strategies, tasks, factors and safety.

**ICP**
: Incident Control Point. The location from which the incident is controlled. "Information collection plan" is always written in full, never abbreviated.

**IMT**
: Incident Management Team.

**IPP**
: Initial Planning Point. The point distances are measured from, usually the LKP or PLS.

**IR**
: Initial Response. The first phase of a search, driven by pre-determined, statistically based actions.

**LKP**
: Last Known Point. The last location the subject is known to have been, from physical evidence.

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**LPB**
: Lost Person Behaviour. Statistical data on how people in different subject categories behave and where they are found.

**MBO**
: Management by Objectives.

**MMISSI TD**
: Mnemonic for the initial response process: missing person report, map, information collection plan, scenario, search urgency, incident action plan, task, debrief.

**MOST**
: Mission, Objectives, Strategies, Tasks. The structure of an IAP.

**MPR**
: Missing Person Report.

**OSC**
: On-Scene Coordinator.

**PLS**
: Point Last Seen. The last location where someone saw the subject.

**POA**
: Probability of Area. The likelihood that the subject is in a given region or segment.

**POD**
: Probability of Detection. The likelihood that a search of an area would have found the subject if they were there.

**POS**
: Probability of Success. POA × POD for a searched area.

**RCCNZ**
: Rescue Coordination Centre New Zealand. Coordinating Authority for Category II operations.

**SAD**
: Search Area Determination. Setting the boundary of the search area.

**SAR**
: Search and Rescue.

**SAREX**
: Search and rescue exercise.

**SAROP**
: Search and rescue operation.

**SEE**
: Search Effort Evaluation. The team debrief that establishes coverage and POD.

**SET**
: Speed, Efficiency, Thoroughness. Tells a team which matters most for its task.

**SMART**
: Specific, Measurable, Achievable, Relevant, Time-bound. Criteria for objectives.

**SMS**
: Safety Management System.

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CHAPTER

SEVEN

# TRACEABILITY STATUS

Generated 2026-10-17 from 44 items: 15 principles, 17 concepts, 6 guidelines, 3 processes, 3 forms.

## 7.1 Needs attention

| Status | Item | Detail |
|---|---|---|
| warning | *PRN-15 Iwi and community partnerships* | no concept derives from this principle yet |
| stale | *CON-02 Risk management and safety* | written against PRN-13 v1; PRN-13 is now v2 |
| overdue | *GDL-04 Rescue phase: known location* | last reviewed 2025-06-30 |

## 7.2 Derivation tree

- *PRN-01 Unified and consistent approach*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-02 Adherence to a national management framework (CIMS)*, v1
  - *CON-04 Notification and awareness*, v1
    - *PRC-03 Notification and on-scene coordination*, v1
  - *CON-06 Incident management teams and CIMS*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-03 Fostering partnerships and relationships*, v1
  - *CON-05 Leadership*, v1
  - *CON-06 Incident management teams and CIMS*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-04 Importance of regional collaboration*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-05 Proactive planning and readiness*, v1
  - *CON-03 Pre-operational readiness plans*, v1
  - *CON-13 Incident action plans*, v1
    - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
      - *FRM-03 IMT summary checklist*, v1

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- *PRN-06 Effective and supportive leadership*, v1
  - *CON-05 Leadership*, v1
- *PRN-07 The critical role of communications*, v1
  - *CON-10 Communications*, v1
- *PRN-08 Evidence-based and intelligence-led operations*, v1
  - *CON-08 Maths in SAR*, v1
  - *CON-09 Lost and missing person behaviour*, v1
  - *CON-11 Evidence-based and intelligence-led operations*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
  - *CON-17 Search theory: POA × POD = POS*, v1
    - *GDL-05 Extended search planning phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-02 Extended search planning cycle*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
        - *FRM-02 Scenario weighting sheet*, v1
    - *GDL-06 Suspension*, v1
      - *FRM-03 IMT summary checklist*, v1
- *PRN-09 Search urgency*, v1
  - *CON-04 Notification and awareness*, v1
    - *PRC-03 Notification and on-scene coordination*, v1
  - *CON-15 The initial response process*, v1
    - *GDL-02 Initial response phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-01 Initial response (MMISSI TD)*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *PRN-10 Structured information management*, v1
  - *CON-11 Evidence-based and intelligence-led operations*, v1
    - *GDL-01 Managing an IMT: key actions*, v1
- *PRN-11 Systematic and methodical approach: POA × POD = POS*, v1
  - *CON-07 Phases of a land search and rescue operation*, v1
    - *GDL-04 Rescue phase: known location*, v1
    - *GDL-06 Suspension*, v1
      - *FRM-03 IMT summary checklist*, v1
  - *CON-08 Maths in SAR*, v1

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- *CON-12 Management by objectives*, v1
  - *GDL-01 Managing an IMT: key actions*, v1
  - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
    - *FRM-03 IMT summary checklist*, v1
- *CON-13 Incident action plans*, v1
  - *GDL-03 Handovers, demobilisation and operational debriefs*, v1
    - *FRM-03 IMT summary checklist*, v1
- *CON-14 The Bike Wheel model*, v1
  - *GDL-02 Initial response phase*, v1
    - *FRM-03 IMT summary checklist*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *CON-15 The initial response process*, v1
  - *GDL-02 Initial response phase*, v1
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  - *PRC-01 Initial response (MMISSI TD)*, v1
    - *FRM-01 Team debrief (search effort evaluation)*, v1
- *CON-16 The extended search planning process*, v1
  - *GDL-05 Extended search planning phase*, v1
    - *FRM-03 IMT summary checklist*, v1
    - *PRC-02 Extended search planning cycle*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
      - *FRM-02 Scenario weighting sheet*, v1
  - *PRC-02 Extended search planning cycle*, v1
    - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *FRM-02 Scenario weighting sheet*, v1
- *CON-17 Search theory: POA × POD = POS*, v1
  - *GDL-05 Extended search planning phase*, v1
    - *FRM-03 IMT summary checklist*, v1
    - *PRC-02 Extended search planning cycle*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
      - *FRM-02 Scenario weighting sheet*, v1
  - *GDL-06 Suspension*, v1
    - *FRM-03 IMT summary checklist*, v1

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- *PRN-12 The principle of simplicity*, v1
  - *CON-15 The initial response process*, v1
    - *GDL-02 Initial response phase*, v1
      - *FRM-03 IMT summary checklist*, v1
      - *PRC-01 Initial response (MMISSI TD)*, v1
        - *FRM-01 Team debrief (search effort evaluation)*, v1
    - *PRC-01 Initial response (MMISSI TD)*, v1
      - *FRM-01 Team debrief (search effort evaluation)*, v1
- *PRN-13 Prioritising safety*, v2
  - *CON-02 Risk management and safety*, v1 (stale)
- *PRN-14 Continuous improvement*, v1
  - *CON-01 Hierarchical framework of the doctrine*, v1
- *PRN-15 Iwi and community partnerships*, v1

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CHAPTER

EIGHT

# HOW THIS DOCTRINE IS MAINTAINED

## 8.1 Two review tiers

| Tier | Folders | To merge a change |
|---|---|---|
| Doctrine | Parts 1, 2, 3.1, 3.2, glossary, shared snippets | Approval from a doctrine owner (sector review group) |
| Guidance | 3.3–3.6, forms | One approval from any reviewer |

Doctrine changes slowly and needs agreement; guidance and forms change often and need light review. The split is set in `.github/CODEOWNERS`.

## 8.2 Items and derivation

Principles, concepts, guidelines, processes and forms are **items**. Each item page starts with a short block of metadata:

```yaml
---
id: GDL-05
type: guideline
title: Extended search planning phase
derives_from:      # parent ID: parent version this was written against
  CON-16: 1
  CON-17: 1
owner: doctrine-group
last_reviewed: 2026-10-01
---
```

Allowed links run down the hierarchy: principle → concept → guideline → process → form. A guideline can't derive straight from a principle.

When an item's meaning changes, its author bumps `version`. Every child written against the old version is then listed as **stale** on the *Traceability status* page until someone checks it and updates its `derives_from` number. The tooling confirms that someone *looked*; it can't judge whether the child is still right.

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## 8.3 Checks on every change

| Check | Result |
|---|---|
| Parent ID doesn't exist, or wrong parent type | Fails |
| Child claims a newer parent version than exists | Fails |
| Item needs children and has none | Warning |
| Child written against an older parent version | Listed as stale |
| Parent text changed but version not bumped | Warning |
| `last_reviewed` older than the review interval | Listed as overdue |

## 8.4 Say it once

- Repeated text, such as the *POD* scale, lives in one snippet file and is included on every page that needs it (five pages here).
- Acronyms are defined once in the *Glossary*.
- The organisation name comes from one setting, so it reads "LandSAR" everywhere.

## 8.5 What the demo content shows

Text is condensed from draft v0.1, numbered as in the draft. Author notes, names and statistics still being verified are left out. Three states are deliberate, so the *Traceability status* page has something to show:

| State | Item | Why |
|---|---|---|
| Stale | *CON-02 Risk management and safety* | Written against PRN-13 v1; PRN-13 is now v2 |
| Coverage gap | *PRN-15 Iwi and community partnerships* | No concept derives from it yet |
| Overdue review | *GDL-04 Rescue phase: known location* | Last reviewed more than a year ago |

## 8.6 Releases

Each release has a version number like `v1.2.0`: edition, amendment and correction. A new edition means doctrine in Parts 1, 2 or 3.1–3.2 changed meaning; an amendment means guidance, processes or forms changed; a correction changes wording only. Release candidates such as `v2.0.0-rc.1` go out for consultation before the final version. Each release publishes the site and a PDF, and the *Changelog* lists what changed and who contributed. Operational reviews cite the version number shown at the top of every page.

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CHAPTER

NINE

# CHANGELOG

## 9.1 v0.2.0-demo

- Condensed content from draft v0.1 for Parts 1, 2, 3.1 and 3.3–3.6: 15 principles, 17 concepts, 6 guidelines, 3 processes and 3 forms, numbered as in the draft.
- Expanded glossary; POD scale shown as drafted, in one shared snippet.
- Demonstration states: stale CON-02, coverage gap PRN-15, overdue GDL-04.

## 9.2 v0.1.0-demo

- Repository structure for Parts 1–3 and forms.
- One traced example thread: evidence-based decisions → POA/POD/POS → recording POD → search period cycle → team debrief form.
- Deliberate demonstration states: one stale item (CON-03), one coverage gap (PRN-04), one overdue review (GDL-01).

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CHAPTER

TEN

# CONTRIBUTORS

Everyone who has contributed to this doctrine. People are added when a release includes a pull request they wrote or co-wrote; add anyone who contributed another way, such as through workshops or review comments, by hand.

- Tobias Schmidt

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CHAPTER

ELEVEN

# LICENCE AND AUTHORITY

Only doctrine published by Land Search and Rescue New Zealand (LandSAR) is authoritative. Check the version on the front page against the latest release before relying on a copy.

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